

II Arcustangens-Funktion

1 Tangens- und Arcustangens-Funktion

1.1 Ableitung der Arcustangens-Funktion

1.2 Aufgaben

Übung II.1 S. 31/2

2. a) $f(x) = \arctan(\sqrt[3]{x-3})$; $u(x) = \sqrt[3]{x-3}$; $v(x) = x-3$

- Bestimmung der Definitionsmenge D_f :

$$v(x) \geq 0 :$$

$$v(x) = 0$$

$$x-3=0 \quad | \quad \sqrt[3]{}$$

$$x_1 = 3 \text{ (e)}$$

$$\Rightarrow \underline{\underline{D_f = [3; \infty[}}$$

	$-\infty < x < 3$	$x = 3$	$3 < x < \infty$
$v(x)$	-	0	+
$u(x)$	n. d.	0	+

- Nullstellen:

$$f(x) = 0 \quad \Rightarrow \quad u(x) = 0$$

$$x_1 = 3 \text{ (e)}; \quad x_1 \in D_f$$

$$\underline{\underline{N_1(3/0)}}$$

b) $f(x) = \arctan(x^2 - 16)$; $u(x) = x^2 - 16$

- Bestimmung der Definitionsmenge D_f :

$$\Rightarrow \underline{\underline{D_f = \mathbb{R}}}$$

- Nullstellen:

$$f(x) = 0 \quad \Rightarrow \quad u(x) = 0$$

$$x^2 - 16 = 0 \quad | \quad +16$$

$$x^2 = 16 \quad | \quad \pm \sqrt[2]{}$$

$$x_1 = -4 \text{ (e)}; \quad x_1 \in D_f \quad x_2 = 4 \text{ (e)}; \quad x_2 \in D_f$$

$$\underline{\underline{N_1(-4/0); \quad N_2(4/0)}}$$

c) $f(x) = \frac{1}{\arctan\left(\frac{1}{x}\right)}$;

$z_1(x) = 1; \quad n_1(x) = \arctan\left(\frac{1}{x}\right); \quad u(x) = \frac{1}{x}; \quad z_2(x) = 1; \quad n_2(x) = x$

- Bestimmung der Definitionsmenge D_f :

$n_1(x) \neq 0$ für $n_2(x) \neq 0$:

$x_1 = 0$ (e)

	$-\infty < x < 0$	$x = 0$	$0 < x < \infty$
$z_2(x)$	1	1	1
$n_2(x)$	–	0	+
$u(x)$	–	n. d.	+
$z_1(x)$	1	1	1
$n_1(x)$	–	0	+
$f(x)$	–	n. d.	+

$\Rightarrow \underline{\underline{D_f = \mathbb{R} \setminus \{0\}}}$

- Nullstellen:

$z_1(x) > 0 \Rightarrow$ keine Nullstellen

d) $f(x) = \arctan\left(\sqrt[2]{4x^2 - 1}\right); \quad u(x) = \sqrt[2]{4x^2 - 1}; \quad v(x) = 4x^2 - 1$

- Bestimmung der Definitionsmenge D_f :

$v(x) \geq 0$:

$v(x) = 0$

$4x^2 - 1 = 0 \quad | \quad +1$

$4x^2 = 1 \quad | \quad :4$

$x^2 = \frac{1}{4} \quad | \quad \sqrt{}$

$x_1 = -\frac{1}{2}$ (e); $x_1 \in D_f; \quad x_2 = \frac{1}{2}$ (e); $x_2 \in D_f$

	$-\infty < x < -\frac{1}{2}$	$x = -\frac{1}{2}$	$-\frac{1}{2} < x < \frac{1}{2}$	$x = \frac{1}{2}$	$\frac{1}{2} < x < \infty$
$v(x)$	+	0	–	0	+
$u(x)$	+	0	n. d.	0	+
$f(x)$	+	0	n. d.	0	+

$\Rightarrow \underline{\underline{D_f = [-\frac{1}{2}; \frac{1}{2}]}}$

- Nullstellen:

$$f(x) = 0 \quad \Rightarrow \quad u(x) = 0 \quad \Rightarrow \quad v(x) = 0$$

$$x_1 = -\frac{1}{2} (e); \quad x_1 \in D_f; \quad x_2 = \frac{1}{2} (e); \quad x_2 \in D_f$$

$$\underline{\underline{N_1 \left(-\frac{1}{2}/0 \right); \quad N_2 \left(\frac{1}{2}/0 \right)}}$$

Übung II.2 S. 32/4

4. a) $f(x) = \arctan(x^2 + 7); \quad D_f = \mathbb{R}$

$$\begin{aligned} f'(x) &= \frac{1}{1 + (x^2 + 7)^2} \cdot 2x \\ &= \frac{2x}{1 + x^4 + 2x^2 \cdot 7 + 49} \\ &= \underline{\underline{\frac{2x}{x^4 + 14x^2 + 50}}} \end{aligned}$$

b) $f(x) = \arctan\left(\sqrt[2]{x + \frac{1}{2}}\right); \quad D_f = \left[-\frac{1}{2}; \infty\right[$

$$\begin{aligned} f'(x) &= \frac{1}{1 + \left(\sqrt[2]{x + \frac{1}{2}}\right)^2} \cdot \frac{1}{2} \cdot \left(x + \frac{1}{2}\right)^{-\frac{1}{2}} \cdot 1 \\ &= \frac{1}{x + \frac{3}{2}} \cdot \frac{1}{2 \cdot \sqrt[2]{x + \frac{1}{2}}} \\ &= \frac{1}{\left(x + \frac{3}{2}\right) \cdot 2 \cdot \sqrt[2]{x + \frac{1}{2}}} \\ &= \underline{\underline{\frac{1}{(2x + 3) \cdot \sqrt[2]{x + \frac{1}{2}}}}} \end{aligned}$$

c) $f(x) = \arctan(e^x); \quad D_f = \mathbb{R}$

$$\begin{aligned} f'(x) &= \frac{1}{1 + (e^x)^2} \cdot e^x \cdot 1 \\ &= \underline{\underline{\frac{e^x}{e^{2x} + 1}}} \end{aligned}$$

$$\text{d) } f(x) = \arctan(x) + \arctan\left(\frac{1}{x}\right); \quad D_f = \mathbb{R} \setminus \{0\}$$

$$f'(x) = \frac{1}{1+(x)^2} \cdot 1 + \frac{1}{1+\left(\frac{1}{x}\right)^2} \cdot (-1) \cdot \frac{1}{x^2}$$

$$= \frac{1}{x^2+1} + (-1) \cdot \frac{1}{1+\frac{1}{x^2}} \cdot \frac{1}{x^2}$$

$$= \frac{1}{x^2+1} - \frac{1}{x^2+1}$$

$$\underline{\underline{= 0}}$$